

Ha Ha! Versus Aha! A Direct Comparison of Humor to Nonhumorous Insight for Determining the Neural Correlates of Mirth

Ori Amir¹, Irving Biederman^{1,2}, Zhuangjun Wang² and Xiaokun Xu¹

¹Department of Psychology, ²Neuroscience Program, University of Southern California, Los Angeles, CA 90089-1061, USA

Address correspondence to Ori Amir, Department of Psychology, University of Southern California, SGM 501, 3620 South McClintock Ave. Los Angeles, CA 90089-1061, USA. Email: oamir@usc.edu

While humor typically involves a surprising discovery, not all discoveries are perceived as humorous or lead to a feeling of mirth. Is there a difference in the neural signature of humorous versus nonhumorous discovery? Subjects viewed drawings that were uninterpretable until a caption was presented that provided either: 1) a nonhumorous interpretation (or insight) of an object from an unusual or partial view (UV) or 2) a humorous interpretation (HU) of the image achieved by linking remote and unexpected concepts. fMRI activation elicited by the UV captions was a subset of that elicited by the humorous HU captions, with only the latter showing activity in the temporal poles and temporo-occipital junction (linking remote concepts), and medial prefrontal cortex (unexpected reward). Mirth may be a consequence of the linking of remote ideas producing high—and unexpected—activation in association and classical reward areas. We suggest that this process is mediated by opioid activity as part of a system rewarding attention to novel information.

Keywords: cortical μ -opioid gradient, fMRI, humor, medial prefrontal cortex, temporal pole

Introduction

Humor involves a discovery of an unexpected interpretation or perspective similar to what occurs when we experience insight. However, not all insight experiences are funny. Previous imaging studies of humor have contrasted humorous stimuli with the same stimuli without the humorous element (e.g., Mobbs et al. 2003; Moran et al. 2004; Bartolo et al. 2006; Kohn et al. 2011), and/or compared different kinds of humor (e.g., Goel and Dolan 2001; Watson et al. 2007; Samson et al. 2008, 2009; Franklin and Adams 2011). Unfortunately, a comparison among subtypes of humor necessarily ignores the regions involved in all humor, and a comparison of humorous stimuli to the same stimuli with the humorous element removed may be too broad, as controls lack the element of discovery. The present investigation directly compares humorous versus nonhumorous cognitive discovery.

More recently studies have compared humor with more subtle controls, for example, humorous clips versus nonhumorous stimuli that inspire a positive/enjoyable feeling (Neely et al. 2012; Vrticka et al. 2013), humorous resolvable incongruities versus nonresolvable incongruities (Samson et al. 2008, 2009; Bekinschtein et al. 2011; Chan, Chou, Chen, Liang 2012; Chan, Chou, Chen, Yeh et al. 2012). Yet in all of those studies, an element of discovery was arguably present in the humorous conditions but not in the controls, or at least, was not systematically controlled for.

The imaging studies of humor cited above generally reported activation in portions of the temporal lobes often near the junctions with the parietal or occipital lobes and/or the

temporal poles (TPs) and some classical reward regions. Additional regions were also reported, but not consistently across the different studies. The inconsistency might be the result of a lack of control for the element of discovery, as well as additional confounds, common in comedy, which enhance the reward, but are not necessary for humor, such as schadenfreude, superiority, or sexual titillation (Hurley et al. 2011). In the present investigation, we use humorous stimuli that were not designed to induce these emotions. In some of the prior experiments, as noted by Bartolo et al. (2006), humor relied heavily on attribution of intentions, eliciting activity in the regions associated with theory of mind. Our stimuli did not rely heavily on such attributions.

Several imaging studies have explored the neural basis of insight (e.g., Bowden et al. 2005; Aziz-Zadeh et al. 2009). Owing to the large variation in the nature of the stimuli, the tasks, and the controls (ranging from anagrams to physical problem solving involving object interactions), no consistent pattern of activation emerged across the various studies (for a review, see Dietrich and Kanso 2010).

To study the neural correlates of mirth (enjoying humor) while controlling the element of discovery, subjects in the present investigation viewed 2 kinds of simple line drawings, that were, by themselves, uninterpretable with respect to their possible referents (Fig. 1). A subsequently presented caption would be provided that elicited, for one condition, an insight-like (but nonhumorous) interpretation of an object depicted from a partial or unusual view (UV) (Fig. 1A,B) by providing the referent for that object (UV condition). Responses to UV stimuli were compared with responses to the second kind of drawing/caption, with a humorous interpretation (HU) (Fig. 1C,D), based on a surprising linking of remote associations (HU condition). Each HU and UV stimulus had a control caption that merely provided a physical description of the drawing, adding no referential information. For each drawing, half the subjects viewed the original caption and half the control caption. Critically, both UV and HU conditions (unlike their controls) result in a discovery, that is, the referent or a new interpretation, of the previously uninterpretable line drawing.

Materials and Methods

Subjects viewed simple line drawings (Fig. 1) whose referent was inaccessible in the absence of a caption. To compare the brain's response with humor and insight, we compared the activation elicited from humorous “doodles” (Price 1955, 1976, 2000), such as the ones shown in Figure 1C,D, in which the humor is based on the unexpected linking of remote concepts, to drawings of objects in partial or unusual views (Nishimoto et al. 2010; e.g., Fig. 1A,B) defining an insight condition where the description provides the referent but does not elicit a humorous response. There is some disagreement in the literature whether the label “insight” should be applied to instances in which the

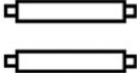
	Unusual View Interpretation (Insight) Drawing:		Humorous Interpretation Drawing:	
Drawing:	A 	B 	C 	D 
Physical Description:	Thick T-junctions	Rectangles with square tabs	Two smaller horizontal ellipses in a larger vertical ellipse.	A plethora of dots surrounded concentrically around a single dot
Interpretive Description:	Trumpet Valves	Fluorescent light bulbs.	Close-up of a pig looking at book titles in a library.	Germes avoiding a friend who caught antibiotics.

Figure 1. Examples of the 4 conditions. Credits: (A and B): Adopted from Nishimoto et al. (2010). (C and D): Doodles excerpts from “Doodles - the Classic Collection” © 2000 by Tallfellow press, Los Angeles. Used by Permission. All rights reserved. (C): Doodle orientation and caption differs from the original Doodle.

solution is provided, as is the case in our experiment, or only when subjects reach the solution on their own. However, in the present experiment providing the solution for the insight condition renders that condition more comparable with the humorous doodles condition, as the HU is similarly provided (rather than asking the subjects to come up with it themselves).

Participants

Fifteen adults, 7 females, all right handed (except for 1 male), age 19–31 (mean = 22.6). All were students at the University of Southern California, except for one graduate from another university. We obtained informed consent from all subjects, and they were compensated for their participation. The study was approved by the University Park Institutional Review Board at the University of Southern California.

Stimuli

The stimuli were 200 line drawings along with both interpretive and physical descriptions. One hundred images whose original captions provided the HU condition were scanned from 3 books by Price (1955, 1976, 2000), whose drawings with their captions he termed “Doodles.” We used only Doodles that were readily understood and did not require knowledge of out of date conventions or objects. The remaining stimuli, 100 (insight) images whose original captions referred to objects depicted from a partial or UV interpretation were taken from Nishimoto et al. (2010). Out of their database of 196 exemplars created for experimental purposes, 100 were selected based on interpretability.

Interpretive descriptions were generally taken as written by the original authors. Some minor changes were made to the descriptions of UVs to make them more accessible to American subjects. Physical descriptions were created which described the drawing physically without conveying its interpretation and which, on average, approximated the length of the drawing’s interpretive description with respect to the number of words.

Subjects viewed both types of drawings, HUs and UVs, followed by either the referential or the control descriptions for a total of 4 conditions. Each drawing’s 2 descriptions were counterbalanced between subjects so that half the subjects saw the control description and half saw the referential description. A particular drawing was viewed in only a single condition by a given subject.

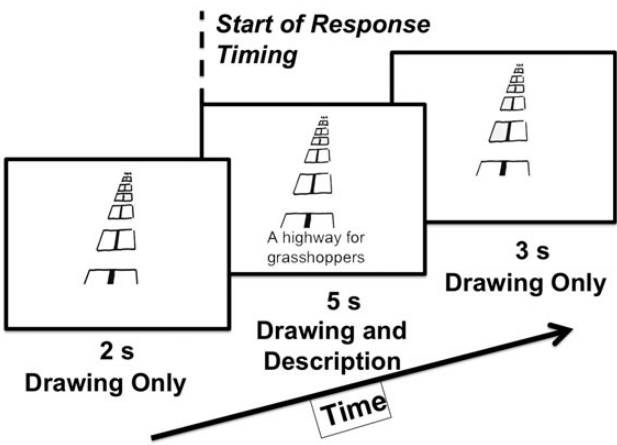


Figure 2. The temporal arrangement of a trial in the fMRI experiment. Doodles excerpts from “Doodles - the Classic Collection” © 2000 by Tallfellow press, Los Angeles. Used by Permission. All rights reserved.

Procedure

Each 10-s trial began with a 2-s display of a drawing (subtending a visual angle of ~8°) in the center of the screen. The drawing was shown alone to pique the subject’s curiosity (Fig. 2). Then for 5 s, either a physical or interpretive description was displayed below the drawing followed by 3 s of just the drawing itself, allowing for the interpretation to be fully appreciated. Trials were separated by a 2-s blank interval. Presentation sequences were programmed with Psychophysics Toolbox (Brainard 1997; Pelli 1997) running on MATLAB (The MathWorks, Natick, MA, USA). Subjects were instructed to rate each drawing for funniness on a scale of 1–4 with 4 buttons in the scanner: 1 indicated they did not understand the description and/or its relationship to the drawing; 2–4 indicated the degree of funniness with 2 being “not funny,” 3 “a little funny,” and 4 “funny.” A pretest had determined that “funny” was more effective than “very funny” in encouraging subjects to use the full scale. Responses were collected from the onset of the drawing to the end of the trial. Note that only one-fourth of the trials were designed to be funny (HU type), while

another one-fourth of the trials were interpretive but of the UVs type, and one-half the trials were the control physical descriptions.

In one session, each subject participated in an anatomical scan and 4 experimental runs of 50 drawings, each lasting 10 min.

Data Acquisition

All fMRI images were scanned at USC’s Dana and David Dornsife Cognitive Neuroscience Imaging Center on a Siemens Trio 3T scanner with a standard 16-channel head coil. Each subject ran in a high-resolution T_1 -weighted structural scan using MPRAGE sequence. (Repetition time (TR) = 1100 ms, 192 sagittal slices, 256×256 matrix size, $1 \times 1 \times 1$ mm voxels).

Functional images were acquired using an echo-planar imaging (EPI) pulse sequence with the parameters: TR = 2000 ms, TE = 30 ms, flip angle = 62° , 256×256 matrix size, in plane resolution 3×3 , 3 mm thick slices, 32 axial slices covering as much of the brain as possible, always including the TPs, but occasionally missing the superior rim of the primary motor and somatosensory cortices.

Data Analysis

Preprocessing (3D motion correction using Trilinear interpolation, 3D spatial smoothing using a 4-mm full-width at half-max Gaussian filter, linear trend removal using a high-pass filter set to 3 cycles over the run’s length) was done with the Brain Voyager software package (Brain Innovation BV, Maastricht, The Netherlands). Statistical analysis was done using MATLAB scripts along with Brain Voyager. Motion corrected functional images were coregistered with the same session’s anatomical scan. Coregistered images were then transformed to Talairach coordinates and underwent statistical analysis.

Statistical analysis was based on a general linear model with a separate regressor for 10 TRs from the beginning of each trial type. The 6 motion correction parameters (3D translation and 3D rotation) were included in the design matrix of the regression to eliminate any potential motion artifacts. We then conducted a whole-brain, random-effects group average analysis. The contrasts between the different conditions were computed as a subtraction of TRs 4–9 (i.e., 7–18 s from the beginning of the trial) of each condition with an uncorrected threshold of $P < 0.005$ (with an additional cluster threshold of 10 consecutive voxels applied). This procedure was used to define the regions showing differential activity for both HU and UV conditions compared with their respective physical controls (HU_{cont} and UV_{cont}), and compared with each other after the controls were subtracted to reveal regions uniquely activated for humor but not insight ([HU – HU_{cont}] – [UV – UV_{cont}]), as well as a conjunction of the 2 conditions to reveal regions activated for both humor and insight ([HU – HU_{cont}] and [UV – UV_{cont}]). In the same way, we also obtained a contrast map of only those HU drawings which were rated 4–“funny” minus the HUs rated 3–“a little funny”, by each subject individually, resulting in a map of regions selective for the degree of funniness. For the rating analysis, all parameters were the same, but we used a fixed effect analysis, because of reduced power (nevertheless, as described in the results, a comparison of the HU with the UV conditions yielded highly similar results to a comparison of a rating of 4 vs. 3, with some additional regions found in the latter analysis). We used the relatively broad time window (TRs 4–9) to make sure that we captured the sum of activations independent of the slight variations over subjects and trials in the time course (e.g., of “getting the joke”).

Results

Behavioral Results

Subjects rated a drawing as “1” (I do not understand) on only 1.3% of the trials, and those trials were removed from further analysis. Mean ratings for the 4 conditions are shown in Table 1. The HU drawings were rated as significantly funnier than the UV interpretation drawings, $t_{(14)} = 12.39$, $P < 0.001$, and the physical description control of the HU condition

Table 1
Mean funniness ratings and SDs for the 4 conditions

Condition	Mean funniness rating	SD
HU (humorous)	3.351	0.336
UV (Insight)	2.220	0.246
HU control	2.090	0.159
UV control	2.035	0.068

Funniness ratings were 2—“not funny,” 3—“a little funny,” and 4—“funny.” Ratings of 1—“I don’t understand” are excluded.

(HU_{cont}), $t_{(14)} = 12.42$, $P < 0.001$. UV drawings, which were not meant to be humorous, still received a slightly, but significantly, higher mean ratings than their UV_{cont}, $t_{(14)} = 2.93$, $P = 0.011$.

Mean response times for responding to the HU drawings was 3.64 s (SD = 0.86), which included the time to read the description. This time was significantly longer than that for UV, $M = 2.92$, $SD = 1.15$, $t_{(14)} = 5.04$, $P < 0.001$. The RTs for the control conditions HU_{cont}, $M = 3.08$, $SD = 1.12$, and UV_{cont}, $M = 2.51$, $SD = 1.18$, were both significantly shorter than those of their respective interpretive conditions, respectively $t_{(14)} = 4.53$ and $t_{(14)} = 7.01$, both P ’s < 0.001 . The 400- to 600-ms RT difference between HU and UV may reflect the longer time required to get the referent for the HU stimuli and, perhaps, also deciding its funniness. Owing to the large window of time selected for the imaging data analysis (TRs 4–9, TR = 2 s), we likely captured all of the event-related activity for both conditions.

Humorous Interpretation Drawings and Unusual View Interpretation Drawings versus Controls

The HU and UV drawings were contrasted with their physical description controls. Figure 3 displays these contrasts on an inflated cortical map (see also Fig. 4, for additional regions not apparent on the inflated maps). We used the $P < 0.005$ threshold for all statistical analysis. We found that the regions exhibiting greater activation for the UVs versus their controls were a subset of those activated for the HUs versus their controls.

A conjunction analysis revealed the regions activated by both UV and HU conditions compared with their respective controls (Table 2). Among those regions were the following: left frontal regions largely overlapping with Broca’s language area, a small portion of the posterior inferior frontal gyrus on the right, previously suggested to involve holding multiple alternative meanings in language processing (Mashal et al. 2005), and an area largely overlapping with bilateral lateral occipital complex (LOC)—a region selective for visual images of object shape (Grill-Spector et al. 1999; Hayworth and Biederman 2006). The parahippocampal gyrus (PHG), a region which activates more strongly to visual scenes compared with single objects (Epstein and Kanwisher 1998), and responds more strongly to visual objects and scenes that produce rich contextual associations (Bar et al. 2008), was activated more for both interpretive conditions.

Increased activation for both interpretive conditions was also observed in the bilateral ventral striatum, and the left amygdala (IAMG) (while the right amygdala was significantly activated for both HU – HU_{cont} and UV – UV_{cont}, no right amygdala activation survived either the conjunction of these 2 contrasts or their subtraction [see Supplementary Tables 1 and 2]),

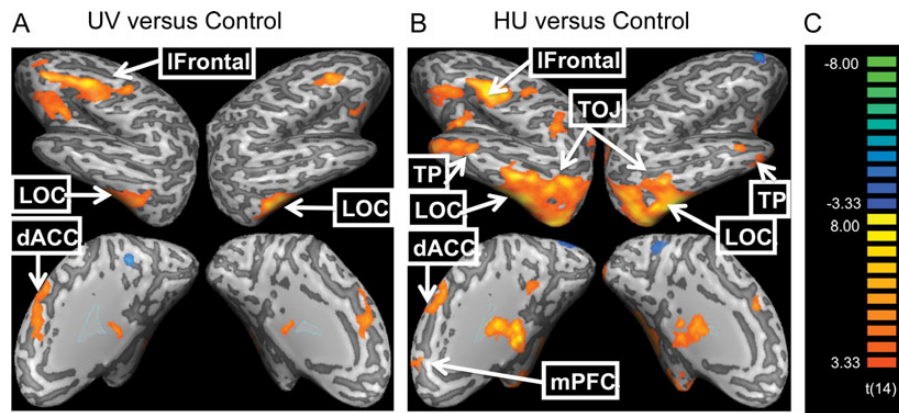


Figure 3. (A) BOLD response to Unusual View (UV) Interpretation "Insight" Drawings minus their physical description controls (UV_{cont}). (B) Humorous Interpretation (HU) Drawings minus their physical description controls (HU_{cont}). (C) Color scale of t -values. Maps are displayed at a threshold of $P < 0.005$ uncorrected. Note: right side of the brain is on the right. TP, temporal pole; LOC, lateral occipital complex; TOJ, temporo-occipital junction; dACC, dorsal anterior cingulate cortex (supplementary motor area); mPFC, medial prefrontal cortex.

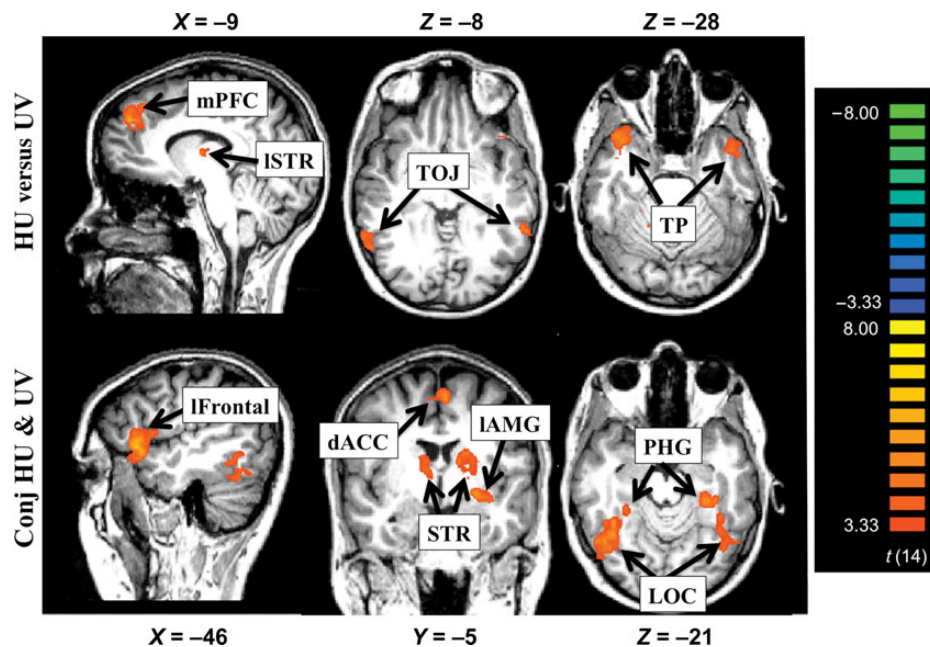


Figure 4. Top—regions activated for HU but not UV (based on the contrast: $[HU - HU_{cont}] - [UV - UV_{cont}]$). Bottom—regions activated for both $HU - HU_{cont}$ and $UV - UV_{cont}$ as revealed by conjunction analysis. All maps are displayed with a threshold of $P < 0.005$, uncorrected. TP, temporal pole; TOJ, temporo-occipital junction; PHG, parahippocampal gyrus; LOC, lateral occipital complex; mPFC, medial prefrontal cortex; IAMG, left amygdala; STR, striatum; dACC, dorsal anterior cingulate cortex (this area of activation includes SMA, supplementary motor area). Figures conform to radiology conventions, with left side on the right.

both of which are part of the dopaminergic reward network (Lee et al. 2004), and both were reported in previous fMRI studies of humor (e.g., IAMG specifically was reported in Mobbs et al. (2003), Bartolo et al. (2006), Watson et al. (2007), although some studies found bilateral amygdala activity, e.g., Moran et al. (2004) and still others reported no amygdala activity). Our findings suggest that activation in these 2 reward regions occurs in response to nonhumorous discovery as well (note, however, that a small region within the left striatum was significantly more activated for HU than UV [see Fig. 4, top left image, and Table 3]). Activity in both the PHG and the striatum has been found to correlate positively with preference for visual scenes (Yue et al. 2007).

Finally, a region extending from the dorsal portions of the anterior cingulate cortex (dACC) to the supplementary motor

area (SMA) was activated for both interpretive conditions. Mobbs et al. (2003) reported activation in this region for humorous stimuli, and they suggested it to be involved in the motor planning of laughter. However, that region's elevated activity for the nonhumorous UV condition suggests an alternative cognitive function: the reassessment of the display based on the new interpretation (an executive function often attributed to ACC, specifically in studies of insight; e.g., Aziz-Zadeh et al. (2009)).

Regions Only Activated for Humorous Interpretations

A subtraction of activation for UV drawings from HU drawings (after each condition's activation was subtracted from its respective control: $[HU - HU_{cont}] - [UV - UV_{cont}]$), revealed 4

Table 2

Regions activated by both HU and UV minus their respective controls, as revealed by a conjunction analysis

ROI	No. of voxels	X	Y	Z	Mean <i>t</i>	Mean <i>P</i> value
rLOC	8571	36	−53	−26	4.090	0.002
lFrontal	5042	−47	20	1	4.071	0.002
rCer	4568	30	−61	−33	4.131	0.002
ISTR	4005	−15	−7	11	3.876	0.002
lLOC	3789	−41	−49	−22	3.772	0.003
ldACC	3736	−5	12	56	3.825	0.002
rSTR	2036	11	−3	11	3.781	0.002
lAMG	1316	−30	−6	−7	3.897	0.002
lPHG	981	−32	−34	−22	3.915	0.002
rPHG	468	28	−40	−24	3.968	0.002
rFrontal	342	55	19	24	3.748	0.003

Note: ROIs are defined using a threshold of $P = 0.005$, uncorrected. ROI sizes in voxels as well as Talairach coordinates, average *t* and *P* values are provided.

l, left; r, right; m, middle; LOC, lateral occipital complex; SMA, supplementary motor cortex; STR, striatum; Cer, cerebellum; PHG, parahippocampal gyrus; AMG, amygdala; dACC, dorsal anterior cingulate cortex; ROI, regions of interest.

Table 3

Regions activated for humor but not nonhumorous discovery (insight) as revealed by the subtraction $[HU - HU_{cont}] - [UV - UV_{cont}]$

ROI	No. of voxels	X	Y	Z	Mean <i>t</i>	Mean <i>P</i> value
rTPJ	4989	38	14	−25	4.113	0.002
rTPJ	3570	42	−82	11	4.437	0.001
lmpFC	3543	−7	48	40	4.118	0.002
ITP	3139	−50	3	−23	3.991	0.002
rTOJ	2545	57	−56	−2	3.876	0.002
rOFC	1133	28	11	−18	3.864	0.002
ITP	1013	−57	−61	6	3.823	0.002
Neg_rParietal	640	17	−47	54	−3.792	0.002
lTOJ	528	−61	−49	−9	3.947	0.002
rCer	515	20	−38	−33	4.037	0.002
rHip	430	25	−32	−16	3.929	0.002
ISTR	375	−7	−11	10	3.795	0.003
Neg_srFrontal	362	26	29	38	−3.838	0.002
Neg_rSTG	311	41	−34	8	−3.702	0.003

Note: ROIs are defined using a threshold of $P = 0.005$, uncorrected. ROI sizes in voxels as well as Talairach coordinates, *t* and *P* values are provided.

l, left; r, right; m, middle; s, superior; Neg, negative (i.e., activation is significantly lower than baseline); TP, temporal pole; TPJ, temporoparietal junction; TOJ, temporo-occipital junction; OFC, orbitofrontal cortex; Cer, cerebellum; mPFC, medial prefrontal cortex; Hip, hippocampus; STR, striatum; STG, superior temporal gyrus; ROI, regions of interest.

regions of activation (see Fig. 4 top and Table 3): 1) The medial prefrontal cortex (mPFC), an area found to be involved in reward learning that responds to unexpected reward contingencies (Rolls 1996), and also has been found to correlate with funniness ratings of stimuli in previous studies (e.g., Goel and Dolan 2001), 2) the TPs, and 3) the temporo-occipital junctions (TOJs) extending to 4) temporoparietal junctions (TPJs). Mobbs et al. (2003) found activation in both TOJ and TP as well, but only on the left side, and suggested TP activity may relate to lexical retrieval or semantic knowledge processing, and the TOJ does semantic processing and may be detecting incongruity/surprise. TPJ activity, particularly in the right hemisphere has been linked to mentalizing; however, the left TPJ is responsive to other similar cognitive tasks not requiring modeling of other minds, that is, perspective taking and comparing interpretations (Perner et al. 2006). Regardless of their exact function in the context of humor, these are association regions involved in high-level semantic processing (Meyer and Damasio 2009; Man et al. 2012). While there is some inconsistency in the humor literature against which we have compared

our results, as discussed in the Introduction, this may be the result of the variable nature of controls (which often either included some funny elements or were lacking the element of discovery). Unlike previous studies, we compared the humorous HU condition with a nonhumorous UV condition which retained the element of discovery, allowing regions responsive to humor to be distinguished from those responsive to insight or reinterpretation.

Eliminating Potential Confounds

The HU Drawings compared with those for the UVs were more complex visually—as there were, on average, more pixels in the drawings, and their verbal interpretation tended to be longer. Consequently, the comparisons of HU and UV interpretive conditions were always made after subtracting from the interpretive conditions their respective descriptive controls. The descriptive controls for the HUs and UVs were such that the images were identical and the captions were of the same length (see Materials and Methods). To assess whether any of the differences between the HUs and the UVs minus their respective controls were the result of differences in figure complexity or descriptive length/content, we examined the contrast between the 2 physical controls: $HU_{cont} - UV_{cont}$. This contrast only yielded significant activation in early visual areas up to and including LOC (but not PHG) and in the left hemisphere frontal language areas (using the same $P < 0.005$, uncorrected, as in all of the analysis we report). Note that none of these regions were the ones differentiating HU from UV activations.

Another concern was that laughter, in the humorous HU condition, would result in head movements that might artificially increase the BOLD signal in some of the regions we reported. While head movements were small to begin with (always < 2 mm, and typically < 0.5 mm), and they were corrected in the preprocessing (see Materials and Methods), we conducted an additional test to confirm that head movements did not explain our findings: we added the 6 motion correction parameters (3D translations and rotations) to the design matrix of the GLM in BrainVoyager, and obtained virtually identical results with exclusion of the motion correction parameters in the design matrix. All results presented here were computed with the design matrix including the motion correction parameters.

Ratings of Humor

We contrasted the activation, within the interpretive HU condition, for drawings that were rated by each subject as 4 (funny), minus those that were rated 3 (a little funny) (Fig. 5) to assess whether a “dose–response” effect would be evident and, if so, whether this effect would also involve those regions differentially activated by the humorous minus the nonhumorous insight conditions. These comparisons included fewer trials (because they constituted only the trials from the HU drawings, and only those rated either 3 or 4). Nevertheless, under the same threshold ($P < 0.005$), robust activation was evident in the same regions that differentiated HU from UV conditions in the previous analyses: bilateral TPs, the temporo-occipital junction, and bilateral medial prefrontal-frontal cortex. However, in this analysis, the region of differential activation-labeled mPFC extended more dorsally within the mPFC, and it was more prominent on the left side. The striatum

Funniness rating 4 versus 3

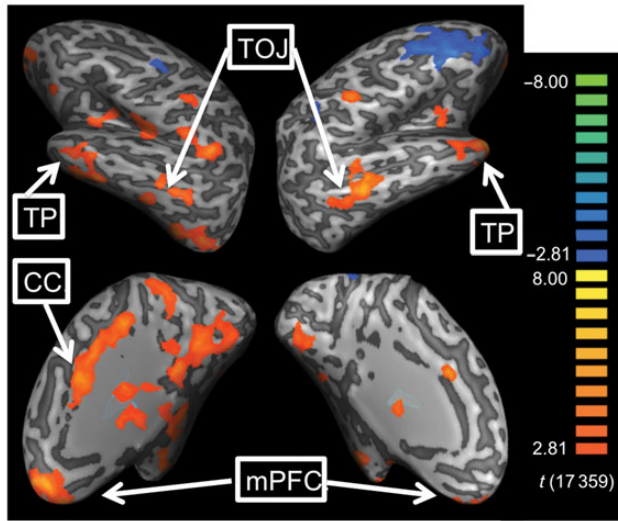


Figure 5. A contrast map of HU drawings rated 4—“Funny” minus those rated 3—“a little funny” for a threshold of $P < 0.005$. mPFC, medial prefrontal cortex; TP, temporal pole; TOJ, temporo-occipital junction; CC, cingulate cortex.

and parahippocampal cortices were also activated bilaterally, as well as most of the left cingulate cortex, precuneus, and lingual gyri on both sides. This overlap between regions activated for higher funniness ratings and the regions specified by comparing the HU minus the UV conditions suggests that the latter difference indeed amounts to a manipulation of humor.

Previous imaging studies of humor that had both a nonhumorous control and obtained funniness ratings did not report that all regions showing higher activation for the humorous condition, relative to the nonhumorous control, also showed a dose response (e.g., Goel and Dolan 2001; Watson et al. 2007). The fact we did find a dose response suggests that we have tighter controls over this complex behavior.

Discussion

The “Humor” Network

Both the humorous HU drawings, and the insight-like UV drawings had an element of discovery and stimulus reinterpretation. However, a network of 4 regions, uniquely and bilaterally active for the humorous stimuli was revealed by subtracting activation for the UV from the HU drawings: the TPs, temporo-occipital junction (TOJ), extending to the temporo-parietal junction (TPJ), and the mPFC. The role of this network in humor processing is further supported by its greater activation in the separate analysis comparing, within the HU condition, those drawings that were rated funniest with those rated as less funny.

What Makes Humor Humor?

In an often cited (almost by default) attempt to characterize the cognitive process underlying humor appreciation, Suls (1972) suggested that humor is a form of problem solving in which there are 2 stages: 1) a perception of incongruity between the “punch line” and what was expected, and 2) a resolution of the incongruity. However, that also characterizes nonhumorous

insight (Ruch and Hehl 1998). What, then, distinguishes the 2 experiences?

Navon (1988) suggested that jokes are characterized by a situation/action that is appropriate from one perspective typically assumed by the protagonist of the joke, but virtually inappropriate or absurd in reality. Similarly, Ruch and Hehl (1998) suggested a third stage to Suls’ theory, in which there is a realization that the resolution does not make sense (is only an “as if” resolution). Hurley et al. (2011) suggested that all humor is based on realizing an inappropriate interpretation/perception was reached because of an erroneous assumption/heuristic. McGraw and Warren (2010) more generally described the conditions for humor as the perception of a benign violation. What is common to the 4 theories above is that some absurd, erroneous, or inappropriate perception/behavior is entertained resulting in violated expectations. Indeed, the funniest drawings elicited activity in the anterior cingulate cortex, which has been suggested to encode errors/conflicts (Botvinick et al. 2004; Brown and Braver 2007; Shenhav et al. 2013). These accounts also imply that, in order for these conditions to be met, humor needs to be based on remote associations—associations not readily elicited by the set up—simply because otherwise it might be impossible to mislead or surprise the listener. The linking of the remote associations (e.g., in Fig. 1C, a pig looking at book titles in the library) likely occurs in association cortex where its novelty, the violation of expectations, and/or the rejection of the (apparently appropriate) link as absurd (e.g., because pigs do not read), necessarily results in heightened activity in association regions if the elements are to be actively maintained so that they can be linked by the punch line (or caption, in our study). Once we hear a joke, its repetition is no longer funny as a conceptual structure has been formed that has already integrated the previous remote associations.

Accordingly, we find that these humorous stimuli induce activation in association regions, namely bilateral TOJ, TPJ, and TP, where information converges, and where remote associations may be integrated.

Could the same explanation apply to any other arbitrary constellation of regions? We believe it could not, since this constellation of regions is unique in satisfying 2 requirements: 1) it consists of higher order association areas where remote associations converge to give rise to semantic interpretations; 2) the regions have been reported in previous independent investigations of humor. (Although those regions have been reported in previous investigations, they generally were reported as part of a larger constellation of regions, and our work reduces the set of relevant regions, by controlling for the element of discovery.)

The Element of Discovery

Unlike previous studies in which only the humorous stimuli retained an element of discovery, both the humorous HU and nonhumorous (insight) UV conditions in our study retained an element of discovery, that is, the drawings were uninterpretable until their descriptions revealed what they depicted. Regions that are activated for the UV condition (minus its descriptive controls) appear to be largely a subset of those activated by the HU condition (minus its controls). Our findings (see The Humor Network), therefore, suggest reconsideration of at least 2 interpretations of prior studies.

Watson et al. (2007) found that, late in the processing of their jokes, sight gags led to an increased activation in visual areas, and verbal gags led to increased activation in language areas. They interpreted the findings as support for Suls' (1972) incongruity-resolution theory of humor, suggesting the late visual or linguistic activity arises from the resolution of the visual or linguistic incongruity, respectively. Goel and Dolan (2001) used similar results to reach the same conclusion. However, we found a late increase in activation of LOC (a visual object recognition region) for both HU and UV conditions compared with their controls (Fig. 6). In other words, we found the same activation pattern of "resolution" for both our humorous HU condition and the nonhumorous UV

condition, results that challenge Suls' characterization of the conditions for humor.

Second, as discussed in the Results and Discussion, since the SMA was activated for both HU and UV, it seems less likely its only function in humor processing is laughter production (as was suggested by Mobbs et al. 2003).

As noted, humor typically has an element of absurdity or benign violations (e.g., Navon 1988). In visual humor, typical depictions might include talking animals or a person in an incongruous pose. That is, even without the caption, such depictions look funny—and would have imposed a serious confound in our effort to distinguish humor from discovery. Fortunately, the HU doodles depicted no such absurdities. Instead, like the control UV stimuli, they appeared "abstract" with no real-world referent. It was the caption, not the drawing, that conveyed the absurdity of the referent that thus rendered the composite funny. We could thus assess the impact of the captions alone, unconfounded with the images.

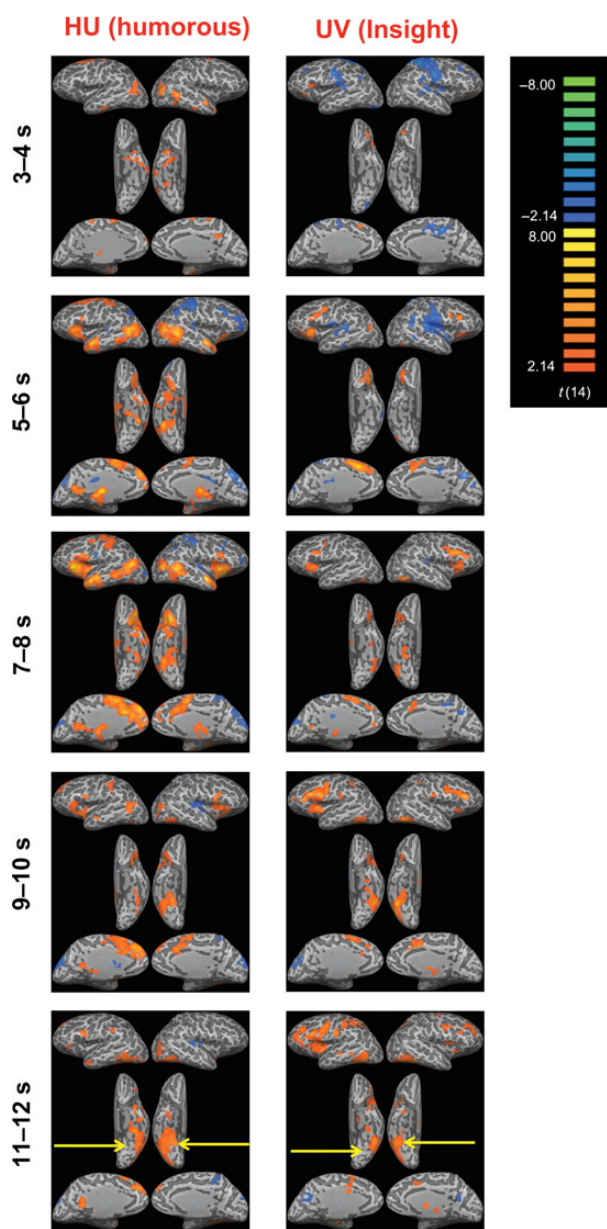


Figure 6. Time course of activation for the HU and UV conditions minus their controls, for $P < 0.05$, uncorrected (the relaxed threshold here is for presentation purposes only, all analysis and other figures used a $P < 0.005$ threshold). Notice that activation in LOC increases (relative to controls) starting around 9 s (or a little earlier) after the beginning of the trial, suggesting a reinterpretation of the drawing.

Why is Humor Pleasurable?

Many theories have suggested evolutionary advantages for enjoying humor, such as, a motivation for error debugging (Hurley et al. 2011) and a number of previous studies reported activation in some classical reward areas (e.g., Mobbs et al. 2003). We are aware of only one, however, that suggested a neural mechanism linking the cognitive processing of humor and the pleasurable feeling of mirth (e.g., Biederman and Vessel 2006).

There is near consensus in the field that "surprise" is an element of humor, and that the surprise is positive (or at least benign). Often the surprise induces a feeling of superiority, Schadenfreude, relief, or sexual titillations, but these factors are not necessary (Hurley et al. 2011). None of these were aspects of our humorous stimuli. The positive surprises induced by our stimuli seem to be the "cleverness" of the interpretation. Indeed, in a separate survey, 20 student subjects rated the same HUs used in the fMRI study, and the survey revealed a high correlation of Funniness ratings with ratings of both Cleverness ($r = 0.823$, $P < 0.001$) and Surprise ($r = 0.711$, $P < 0.001$) but not Complexity ($r = -0.122$, $P = 0.188$). The link between cleverness and pleasure may be an aspect of a general motivational system, which renders the consumption of novel and richly interpretable information as pleasurable.

Lewis et al. (1981) discovered a gradient of μ -opioid receptors in the ventral cortical visual pathway, sparse in early sensory areas, but gradually increasing in density through higher sensory cortices, with the greatest density in association cortex (e.g., PHG, TPs). Zadina et al. (1997) reported the presence of μ -opioid receptors in prefrontal cortex. Biederman and Vessel (2006) hypothesized that neural activation in association cortex would be greater in response to richly interpretable, novel, and surprising experiences, resulting in greater opioid release, possibly triggering further activation of classical reward regions, producing pleasure. They proposed that such activation of cortical opioids might be the neural basis for the human motivation for seeking novel, richly interpretable information (for additional evidence, see Yue et al. 2007; Amir et al. 2011).

The TPs and TOJs association areas were significantly activated for HUs but not UVs. Also responding exclusively to HUs, the mPFC, which is implicated in reward learning and

responds most strongly to unexpected rewards (or their absence; [Rolls 1996](#); [Lee et al. 2004](#)). This characterization fits well with the unexpected and rewarding nature of punch lines.

In conclusion, the mirthful feeling accompanying humor appreciation may come about as a result of unexpected greater activation in cortical association regions rich in μ -opioid receptors (e.g., bilateral TOJ and TP), triggering, in response to this unexpected reward, activation in classical reward/reward-learning regions such as the mPFC, I AMG, and striatum (with the latter 2 regions also activated for nonhumorous discovery).

Summary

A handful of regions differentiate the discovery inherent in humor from nonhumorous discovery. The same regions were also more responsive to funnier humorous stimuli, thus exhibiting a dose–response effect. We hypothesize that the surprisingly high activation in association regions (e.g., TPs and TOJ) leads to the high activation in reward regions (e.g., mPFC) in line with previous studies (e.g., [Biederman and Vessel 2006](#)) that linked greater associative activity with perceptual pleasure.

Supplementary Material

Supplementary material can be found at: <http://www.cercor.oxfordjournals.org/>

Funding

This work was supported by National Science Foundation BCS 04-20794, 05-31177, and 06-17699 to I.B.

Notes

We thank Mark Lescroart for sharing his Matlab code, and Jianchang Zhuang for his assistance with the MRI scanning. *Conflict of Interest*: None declared.

References

- Amir O, Biederman I, Hayworth KJ. 2011. The neural basis for shape preferences. *Vision Res.* 51:2198–2206.
- Aziz-Zadeh L, Kaplan J, Iacoboni M. 2009. “Aha!”: the neural correlates of verbal insight solutions. *Hum Brain Mapp.* 30:908–916.
- Bar M, Aminoff E, Schacter D. 2008. Scenes unseen: the parahippocampal cortex intrinsically subserves contextual associations, not scenes or places per se. *J Neurosci.* 28:8539–8544.
- Bartolo A, Benuzzi F, Nocetti L, Baraldi P, Nichelli P. 2006. Humor comprehension and appreciation: an fMRI study. *J Cogn Neurosci.* 18:1789–1798.
- Bekinschtein TA, Davis MH, Rodd JM, Owen AM. 2011. Why clowns taste funny: the relationship between humor and semantic ambiguity. *J Neurosci.* 31:9665–9671.
- Biederman I, Vessel EA. 2006. Perceptual pleasure and the brain. *Am Sci.* 94:249–255.
- Botvinick MM, Cohen JD, Carter CS. 2004. Conflict monitoring and anterior cingulate cortex: an update. *Trends Cogn Sci.* 8:539–546.
- Bowden EM, Jung-Beeman M, Fleck J, Kounios J. 2005. New approaches to demystifying insight. *Trends Cogn Sci.* 9:322–328.
- Brainard DH. 1997. The psychophysics toolbox. *Spat Vis.* 10:433–436.
- Brown JW, Braver TS. 2007. Risk prediction and aversion by anterior cingulate cortex. *Cogn Affect Behav Neurosci.* 7:266–277.
- Chan YC, Chou TL, Chen HC, Liang KC. 2012. Segregating the comprehension and elaboration processing of verbal jokes: an fMRI study. *NeuroImage.* 61:899–906.
- Chan YC, Chou TL, Chen HC, Yeh YC, Lavalley JP, Liang KC, Chang KE. 2012. Towards a neural circuit model of verbal humor processing:

- an fMRI study of the neural substrates of incongruity detection and resolution. *Neuroimage.* 66:169–176.
- Dietrich A, Kanso R. 2010. A review of EEG, ERP, and neuroimaging studies of creativity and insight. *Psychol Bull.* 136:822–848.
- Epstein R, Kanwisher N. 1998. A cortical representation of the local visual environment. *Nature.* 392:598–601.
- Franklin RG Jr, Adams RB Jr. 2011. The reward of a good joke: neural correlates of viewing dynamic displays of stand-up comedy. *Cogn Affect Behav Neurosci.* 11:508–515.
- Goel V, Dolan R. 2001. The functional anatomy of humor: segregating cognitive and affective components. *Nat Neurosci.* 4:237–238.
- Grill-Spector K, Kushnir T, Edelman S, Avidan G, Itzhak Y, Malach R. 1999. Differential processing of objects under various viewing conditions in the human lateral occipital complex. *Neuron.* 24:187–203.
- Hayworth K, Biederman I. 2006. Neural evidence for intermediate representations in object recognition. *Vision Res.* 46:4024–4031.
- Hurley MM, Dennett DC, Adams RB Jr. 2011. Inside jokes: using humor to reverse-engineer the mind. Cambridge, MA: MIT Press.
- Kohn N, Kellermann T, Gur RC, Schneider F, Habel U. 2011. Gender differences in the neural correlates of humor processing: implications for different processing modes. *Neuropsychologia.* 49:888–897.
- Lee GP, Meador KJ, Loring DW, Allison JD, Brown WS, Paul LK, Pillai JJ, Lavin TB. 2004. Neural substrates of emotion as revealed by functional magnetic resonance imaging. *Cogn Behav Neurol.* 17:9–17.
- Lewis ME, Mishkin M, Bragin E, Brown RM, Pert CB, Pert A. 1981. Opiate receptor gradients in monkey cerebral cortex: correspondence with sensory processing hierarchies. *Science.* 211:1166–1169.
- Man K, Kaplan JT, Damasio A, Meyer K. 2012. Sight and sound converge to form modality-invariant representations in temporoparietal cortex. *J Neurosci.* 32:16629–16636.
- Mashal N, Faust M, Hendler T. 2005. The role of the right hemisphere in processing nonsalient metaphorical meanings: application of principal components analysis to fMRI data. *Neuropsychologia.* 43:2084–2100.
- McGraw AP, Warren C. 2010. Benign violations making immoral behavior funny. *Psychol Sci.* 21:1141–1149.
- Meyer K, Damasio A. 2009. Convergence and divergence in a neural architecture for recognition and memory. *Trends Neurosci.* 32:376–382.
- Mobbs D, Greicius M, Abdel-Azim E, Menon V, Reiss A. 2003. Humor modulates the mesolimbic reward centers. *Neuron.* 40:1041–1048.
- Moran J, Wig G, Adams R, Janata P, Kelley W. 2004. Neural correlates of humor detection and appreciation. *NeuroImage.* 21:1055–1060.
- Navon D. 1988. The seemingly appropriate but virtually inappropriate: notes on characteristics of jokes. *Poetics.* 17:207–219.
- Neely MN, Walter E, Black JM, Reiss AL. 2012. Neural correlates of humor detection and appreciation in children. *J Neurosci.* 32:1784–1790.
- Nishimoto T, Ueda T, Miyawaki K, Une Y, Takahashi M. 2010. A normative set of 98 pairs of nonsensical pictures (doodles). *Behav Res Methods.* 42:685–691.
- Pelli DG. 1997. The VideoToolbox software for visual psychophysics: transforming numbers into movies. *Spat Vis.* 10:437–442.
- Perner J, Aichhorn M, Kronbichler M, Staffen W, Ladurner G. 2006. Thinking of mental and other representations: the roles of left and right temporo-parietal junction. *Soc Neurosci.* 1:245–258.
- Price R. 1976. Doodles #1. Los Angeles: Price/Stern/Sloan Publishers Inc.
- Price R. 2000. Doodles: the classic collection. Los Angeles: Tallfellow.
- Price R. 1955. Oodles of doodles. New York: H. Wolff Book.
- Rolls E. 1996. The orbitofrontal cortex. *Philos Trans R Soc Lond B Biol Sci.* 351:1433–1444.
- Ruch W, Hehl FJ. 1998. A two-mode model of humor appreciation: its relation to aesthetic appreciation and simplicity–complexity of personality. In: Ruch W, editor. The sense of humor: explorations of a personality characteristic. Berlin: Mouton de Gruyter. p. 109–142.
- Samson A, Hempelmann C, Huber O, Zysset S. 2009. Neural substrates of incongruity-resolution and nonsense humor. *Neuropsychologia.* 47:1023–1033.

- Samson AC, Zysset S, Huber O. 2008. Cognitive humor processing: different logical mechanisms in nonverbal cartoons—an fMRI study. *Soc Neurosci*. 3:125–140.
- Shenhav A, Botvinick MM, Cohen JD. 2013. The expected value of control: an integrative theory of anterior cingulate cortex function. *Neuron*. 79:217–240.
- Suls J. 1972. A two stage model for the appreciation of jokes and cartoons. In: Goldstein J, McGhee P, editors. *Psychology of humor*. New York: Academic Press. p. 39–57.
- Vrticka P, Neely M, Walter Shelly E, Black JM, Reiss AL. 2013. Sex differences during humor appreciation in child-sibling pairs. *Soc Neurosci*. 8:291–304.
- Watson K, Matthews B, Allman J. 2007. Brain activation during sight gags and language-dependent humor. *Cereb Cortex*. 17:314–324.
- Yue X, Vessel EA, Biederman I. 2007. The neural basis of scene preferences. *NeuroReport*. 18:525–529.
- Zadina JE, Hackler L, Ge LJ, Kastin AJ. 1997. A potent and selective endogenous agonist for the μ -opiate receptor. *Nature*. 386:499–502.